Fixed-point lemma: every one-variable formula has a Q-provable diagonal fixed point

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Editorial note. The equation labels repdiag1 and repdiag2 inside the proof span are internal labels of the counted span.

Editorial scope. One original lemma of the unit's fixed-point section, counted once. The counted lemma and its author proof are retained in full; the closing digression comparing with the computability version is not counted.

Allowed standard prerequisites (editorial).

- The diagonal function diag mapping the Godel number of a formula with one free variable to the Godel number of its diagonalization, and its representing formula D_diag in Q, with the two representability facts established at the start of the unit's fixed-point section: Q derives the instance relating the diagonalization input to the output, and Q derives the universal statement forcing any witnessed y to equal the Godel number of the diagonalization.

Lemma 0.1. *Let $B(x)$ be any formula with one free variable x. Then there is a sentence A such that* $\mathrm{Th}Q \vdash A \leftrightarrow B(\mathrm{gn}\, A)$.

Proof. Given $B(x)$, let $E(x)$ be the formula $\exists y\, (D_{\mathrm{diag}}(x, y) \wedge B(y))$ and let A be its diagonalization, i.e., the formula $E(\mathrm{gn}\, E(x))$.

Since D_{diag} represents diag, and $\mathrm{diag}(\mathrm{gn}\, E(x)) = \mathrm{gn}\, A$, Th$Q$ can derive

$$D_{\mathrm{diag}}(\mathrm{gn}\, E(x), \mathrm{gn}\, A) \tag{1}$$

$$\forall y\, (D_{\mathrm{diag}}(\mathrm{gn}\, E(x), y) \to y = \mathrm{gn}\, A). \tag{2}$$

Now we show that $\mathrm{Th}Q \vdash A \leftrightarrow B(\mathrm{gn}\, A)$. We argue informally, using just logic and facts derivable in ThQ.

First, suppose A, i.e., $E(\mathrm{gn}\, E(x))$. Going back to the definition of $E(x)$, we see that $E(\mathrm{gn}\, E(x))$ just is

$$\exists y\, (D_{\mathrm{diag}}(\mathrm{gn}\, E(x), y) \wedge B(y)).$$

Consider such a y. Since $D_{\mathrm{diag}}(\mathrm{gn}\, E(x), y)$, by 2 , $y = \mathrm{gn}\, A$. So, from $B(y)$ we have $B(\mathrm{gn}\, A)$.

Now suppose $B(\mathrm{gn}\, A)$. By 1 , we have

$$D_{\mathrm{diag}}(\mathrm{gn}\, E(x), \mathrm{gn}\, A) \wedge B(\mathrm{gn}\, A).$$

It follows that

$$\exists y\,(D_{\mathrm{diag}}(\mathrm{gn}\,E(x), y) \wedge B(y)).$$

But that's just $E(\mathrm{gn}\,E(x))$, i.e., A. □